

1. Administrative & Study Identification

Full Study Title: A Longitudinal, Open-label, Multi-center Study to Explore the Relationship Between Signs, Symptoms, Molecular and Imaging Biomarkers in Participants With Dry Eye Disease Following Anti-inflammatory Treatment **Short Title/Acronym:** DED Biomarker and Anti-inflammatory Treatment Study **Protocol Number:** BP45931 **NCT Number:** NCT07025811 **Sponsor/Company Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Vevye® (cyclosporine / ciclosporin ophthalmic solution) and Xiidra® (lifitegrast ophthalmic solution) **Phase of Development:** Phase 4 **Trial Status:** Recruiting **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the performance of biomarkers and their responsiveness to standard-of-care treatments (Vevye® or Xiidra®) in participants with dry eye disease (DED) compared to healthy volunteers (control participants). **Secondary Objectives:** To evaluate changes in DED-related symptoms, visible signs, molecular/imaging markers, tear production, and eye comfort.

2.2 Background & Rationale Dry eye disease (DED) is a condition where eyes do not stay properly hydrated, leading to discomfort and vision problems. While there are FDA-approved treatments available, there is a medical necessity to better understand how these treatments work and what makes healthy eyes different from eyes affected by DED. By tracking patient symptoms alongside molecular and imaging biomarkers, researchers aim to characterize longitudinal changes in signs, symptoms, molecular, and imaging markers of the ocular surface to investigate pathological mechanisms and biomarkers of DED after treatment.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Longitudinal, Multi-center) **Control Type:** Healthy Volunteer Control and Active Comparator (Vevye® vs. Xiidra®) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 350 participants **Number of Sites:** Multi-center **Estimated Study Start:** [DD-MMM-YYYY] **Estimated Primary Completion:** [DD-MMM-YYYY]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria A. Inclusion Criteria for Participants with DED: 1. The participant reported dry eye-related ocular symptoms for at least 6 months before the screening visit. 2. The participant must have either used lubricant eye drops regularly OR had the desire to use lubricant eye drops in the past 3 months. 3. Participant must be classified as having moderate/severe DED. 4. The participant is deemed by the physician to require a prescription medication for DED (e.g., cyclosporin A or lifitegrast).

B. Inclusion Criteria for Healthy Volunteers: 1. The participant does not have overt clinical signs of DED in either eye.

4.2 Exclusion Criteria 1. Intraocular pressure (IOP) >21.00 millimeters of mercury (mmHg) in either eye. 2. Acute allergic conjunctivitis in either eye within 3 months prior to screening. 3. Use of contact lenses within 30 days prior to screening. 4. Punctal plugs within 3 months prior to screening or any history of punctal cauterization or occlusion by an approach different from punctal plugs. 5. Use of ocular anti-inflammatory agents or ocular immunosuppressive agents within 3 months prior to screening. 6. Any intraocular injections (e.g., intravitreal [IVT] anti-vascular endothelial growth factor [VEGF]) within 3 months prior to screening, or such injections planned for within the study period. 7. Any intraocular immunosuppressive implants within 12 months prior to screening. 8. Any history of isotretinoin use within 12 months prior to screening. 9. Uncontrolled ocular or systemic disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Vevye® or Xiidra® eye drops twice daily. Healthy volunteers will receive no treatment. **Route of Administration:** Topical (Ophthalmic solution). **Duration of Treatment:** 12 weeks. **ATC Codes:** S01XA18, L04AD01 (ciclosporin)

5.2 Concomitant & Prohibited Therapy Permitted: Lubricating eye drops (for eligible DED participants). **Prohibited:** Ocular anti-inflammatory or immunosuppressive agents during the study duration.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Ocular Examination
Baseline	Day 0	Initiation of twice-daily eye drops, Symptom Assessment, Cell sampling via EyePrim™
Interim	Week [X]	Safety Assessment

Symptom Check, Comfort Level tracking | | **End of Study** | Week 12
| Final corneal staining, Biomarker cell collection, Exit
Interview |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change From Baseline in Corneal Staining as Measured by Sodium Fluorescein. **Measurement Method:** Clinical Ocular Examination / Corneal Staining.

7.2 Secondary & Exploratory Endpoints * Changes in DED-related symptoms, condition of the conjunctiva, tear production, and eye comfort from baseline to 12 weeks. * Evaluation of biological differences between healthy eyes and DED-affected eyes utilizing molecular and imaging biomarkers derived from cell samples.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical exams and patient reports (including evaluating known side effects such as eye discomfort, temporary blurry vision, and taste changes). **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary effectiveness; Safety Set for AE/SAE evaluations. **Sample Size Justification:** $N=350$ participants appropriately powered to detect biomarker differences and treatment-driven changes in corneal staining. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) or Multiple Imputation for continuous ocular metrics.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard onsite and remote monitoring visits to ensure safety and protocol fidelity. **Audit Trail:** Details on eCRF locking and data clarification forms.

11. Study Identifiers & Governance

Primary ID: BP45931 **Posting ID:** 2846489511269903 **Registry Number:** NCT07025811 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche)
Study Title: A Study to Explore Signs, Symptoms, and Biomarkers in Dry Eye Disease Participants Following Anti-inflammatory Treatment
Official Regulatory Title: A Longitudinal, Open-label, Multi-center Study to Explore the Relationship Between Signs, Symptoms, Molecular and Imaging Biomarkers in Participants With Dry Eye Disease Following Anti-inflammatory Treatment

12. Clinical Framework (Dry Eye Disease Specific)

Disease Area: Dry Eye Disease **UMLS Name:** Dry Eye Disease (Preferred Name: Dry Eye Syndromes) **MeSH Code:** D015352 **SNOMED ID:** 46152009 **HPO Code:** HP:0001097 **Diagnosis Threshold:** Moderate to severe dry eye symptoms lasting at least 6 months
Medical Methodology: Anti-inflammatory ophthalmic treatment (Vevye® or Xiidra®) **Optimization Target:** Improvement in corneal staining parameters and reduction in DED symptoms at 12 weeks

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway utilizing FDA-approved topical treatments
Education Protocol (HCP): Standard study protocol and EyePrim™ cell collection training
Education Protocol (Patient): Ophthalmic drop administration instructions and symptom reporting
Quality Audit Mechanism: HCP perception tracking and patient clinical symptom logging

14. Observation & Follow-Up Schedule

Total Duration: 12 weeks **Onsite Visit Cadence:** Screening, Baseline, and Week 12
Remote Monitoring Frequency: Interim follow-ups as required by protocol
Checklist Review Interval: Symptom reviews periodically (e.g., Q4W)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				